



UNIVERSITY OF PUNJAB

B.S. 4 Years Program / Eighth Semester - Spring 2024

Paper: Analytical Chemistry (Sp. Theory-I)

Course Code: CHEM-431

Roll No.

046858

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

- i. ✓ What are basic mechanism of mass transport in voltammetry?
- ii. ✓ Differentiate between diffusion current and migration current.
- iii. ✓ Briefly explain cell constant.
- iv. ✓ What is role of convection on Faradaic current in Voltametric measurement?
- v. ✓ Define equivalent conductance.
- vi. ✓ Describe two disadvantages of amperometry.
- vii. ✓ What change in conductance when weak acid reacts with strong base?
- viii. ✓ Why mercury electrode is widely used?
- ix. ✓ What is electrical spark?
- x. ✓ Discuss characteristics of Arc sources.
- xi. ✓ How potentiometric titration is useful for determination of indicator for endpoint of titration?
- xii. ✓ Discuss programmable automation.
- xiii. ✓ What is voltammogram?
- xiv. ✓ How Voltametric techniques are classified?
- xv. ✓ Define migration current.

Answer the following questions.

(3x10=30)

- Q.2** (a) ✓ Explain in detail anodic stripping voltammetry.
(b) ✓ Write down application of voltammetry.
- Q.3** (a) ✓ Explain in detail four types of potentiometric titration.
(b) ✓ Describe applications, advantages and disadvantages of amperometer.
- Q.4** (a) ✓ Describe the instrumentation of electric arc/spark.
(b) ✓ Explain half wave potential with suitable example.



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Q.1. Answer the following short questions.

(15x2=30)

- i. What are the advantages of polarography?.
- ii. What is effect of dilution of solution on specific conductance?.
- iii. Write down the at least Four advantages of DME.
- iv. What are sources of Electric Arc?
- v. Describe auxiliary electrode
- vi. Why electroanalytical techniques are superior than analytical methods.
- vii. Write down the limitations of Amperometry.
- viii. What do you mean by diffusion current in voltametry.
- ix. Describe the migration as a mode of mass transport in voltametry.?
- x. Differentiate between migration current and residual current.
- xi. How sputtering takes place at in Glow discharge?
- xii. What do you mean by cyclic voltammetry.
- xiii. What are polarizable and non-polarizable electrodes?
- xiv. What is specific Conductance?
- xv. How conductometry used in precipitation titrations?

Answer the following questions.

(3x10=30)

Q.2(a) Discuss the types of electrodes in Polarography.

(b)-. Write a note on Automation.

Q.3(a) Discuss anodic stripping voltametry.

(b) Discuss different kinds of currents contributing to polarographic waves.

Q.4(a) Discuss biosensors using amperometric transducers

(b)- Write a note on conductometric titrations of strong acid with strong base as well weak base.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2022

Paper: Analytical Chemistry (Sp. Theory-I)

Course Code: CHEM-431

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions. (15x2=30)

- I. What are three basic mechanisms of mass transport in voltammetry?
- II. Differentiate between cathodic current and anodic current in polarography.
- III. Describe the role of inert supporting electrolyte in polarography.
- IV. How sputtering takes place in glow discharge?
- V. Automation possesses some disadvantages. Describe them.
- VI. Write down the principle of amperometry.
- VII. How glow discharge atomization takes place?
- VIII. Distinguish between fixed automation and flexible automation.
- IX. How conductance is measured?
- X. Write down the basic principle of conductometric titrations.
- XI. Give disadvantages of anodic stripping voltammetry.
- XII. Distinguish between migration current and residual current.
- XIII. Describe amperometric titrations involving electro-reducible vs electro-oxidizable species.
- XIV. What is difference between arc and spark?
- XV. What is the effect of charge and size of ions on conductivity?

Q.2. Write detailed answers to the following questions. (6x5=30)

- I. Discuss arc and spark sources.
- II. Explain how conductometric titrations are applied in complexometric and participations reactions.
- III. Discuss instrumentation in amperometric titrations.
- IV. Discuss the construction and working of voltametric cell.
- V. Discuss in detail the factors affecting the limiting current.
- VI. Polarography is applied to organic and inorganic substances. Explain it details.

SHORT QUESTIONS.

1. Write down any two advantages of amperometry?
2. What do you know about residual current?
3. Distinguish between Faradaic current and non-Faradaic current?
4. Briefly describe the factors affecting the capillary characteristics in polarography?
5. Differentiate between Kinetic current and catalytic current?
6. Describe the sputtering with respect to glow discharge?
7. How temperature affects conductance of solution?
8. What is the basic principle of anodic stripping voltammetry?
9. Differentiate between molar and specific conductance?
10. Briefly describe spark ablation?

LONG QUESTIONS.

1. Discuss factors affecting diffusion current?
2. Discuss the instrumentation in anodic stripping voltammetry?
3. Explain the construction and working of DME along with labelled diagram?
4. Discuss different types of titration curves obtained in amperometry giving Examples?
5. Discuss conductometric titration of mixture of strong and weak acid with strong base?
6. Discuss the two types of automated analysis?



UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program / Eighth Semester – 2020

Roll No.

Paper: Analytical Chemistry (Sp. Theory-I)
Course Code: CHEM-431 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers to the following questions. (10x2=20)

- i - What is the effect of dilution on Specific conductance? Give reason also.
- ii- What is the basic principle of Anodic stripping voltammetry?
- iii- What is Ilkovich equation? Give its significance.
- iv- Give application of conductometry in precipitation titrations?
- v- Give difference between molar conductance and specific conductance?
- vi- What do you mean by cyclic voltammetry?
- vii- Give the changes in conductance in the titration of mixture of HCl and acetic acid with NaOH?
- viii- What is cell constant? What is relation between conductivity of cell and cell constant?
- ix- What different arc and spark sources used in spectroscopy?
- x- Give three advantages of DME?

Answers to the following questions. (3x10=30)

- Q.3 (a) Discuss quantitative applications of anodic stripping voltammetry. (5)**
- (b) Write a note on conductometric titrations of strong acid with NaOH and NH_4OH . (5)**
- Q.4 (a) Discuss electrodeposition step and voltammetric completion step in voltammetry. (5)**
- (b) Discuss half wave potential. What information is obtained from polarographic curve? (5)**
- Q.5- (a) Write a note on spark sources. (5)**
- (b) Write note on Qualitative aspect of polarographic analysis (5)**



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – 2019

Roll No.

Paper: Analytical Chemistry (Sp. Theory-I)

Course Code: CHEM-431 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Section I

Q.2- Attempt all Short questions (2x10=20)

i – What is the effect of temperature on conductance of solution?

ii-. How sputtering takes place in glow discharge?

iii-. Give the differences between arc and spark.

iv-How oxygen interferes in polarographic analysis ?

v-Distinguish between specific conductance and molar conductance.

vi- What do you know about residual current in polarography?

vii- What are the advantages and disadvantages of conductometric titrations?

viii Give the characteristic of supporting electrolyte to be used in polarography.

ix- Write down the basic principle of glow discharge technique?

x-. What are the limitations of Amperometry ?.

Section II

Attempt all questions

Q.3(a)-write down the principle and instrumentation of electrical arcs. (5)

(b)-. What is meant by diffusion current ? Discuss factors affecting it. (5)

Q.(4)-(a). Explain conductometric titrations involving

(i) Strong acid with strong base

(ii) Weak acid with strong base (5)

(b) Discuss the applications amperometry? (5)

Q.5- (a) Discuss anodic stripping voltametry. (5)

(b)- . Discuss various types of electrodes in polarography. (5)

ANALYTICAL CHEMISTRY
(SPECIAL THEORY 1)

2018

SHORT QUESTIONS

- Q1-**What is the effect of concentration of ionic size on conduction?
- Q2-** Distinguish between automatic devices and automated devices?
- Q3-** Distinguish between limiting current and diffusion current?
- Q4-** Give any two advantages of amperometry?
- Q5-** What do you mean by residual current?
- Q6-** What are the capillary characteristics and give factors affecting them?
- Q7-** Give difference between continuous analyzer and discrete analyzer?
- Q8-** What are the advantages of conductometric titrations?
- Q9-** What is the role of supporting electrolyte in voltammetry?
- Q10-** What do you know about electric arc?

LONG QUESTIONS

- Q1-** Discuss the analytical applications of conductometry?
- Q2-** What is diffusion current in polarography? Discuss factors affecting it?
- Q3-** Explain different types of titration curves obtained in amperometry?
- Q4-** Write a note on electric arc?
- Q5-** Discuss the influence of Kinetics of Electron Transfer and mass transport on Faradaic current?
- Q6-** Discuss amperometric biosensors?



UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Analytical Chemistry (Sp. Theory-I)
Course Code: CHEM-431

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Section I

Q.2- Attempt all Short questions (2×10=20)

- (i) – What is meant by arc and spark ablation?
- (ii) – What are the advantages of amperometric titrations?
- (iii) – What do you know about polarographic maxima.
- (iv) – How oxygen interferes in polarographic analysis? How the problem is overcome?
- (v) How sputtering takes place in glow discharge cells?
- (vi) Describe briefly the two steps in stripping voltametry?
- (vii) – Describe the two factors that contribute to the electrochemical reactions in polarography.
- (viii) What are the advantages of conductometric titrations over ordinary titrations?
- (ix) How conductance of solution changes under the influence of temperature and concentration of ions?
- (x) Give difference between Specific conductance and molar conductance.

Section II

Attempt all questions

Q.3(a)-. Explain the following conductometric titrations

(i) Mixture of strong acid and weak acid with strong base

(ii) Precipitation titration.. (5)

(b)- Discuss the types of amperometric electrode system. (5)

Q.(4)-(a) Explain instrumentation in polarography. (5)

(b). Discuss glow discharge mechanism and give its applications in chemistry (5)

Q.5- (a) Explain How conductance is measured? (5)

(b)- Discuss the applications of anodic stripping votametry (5)

ANALYTICAL CHEMISTRY

(SPECIAL THEORY 1)

2016

SHORT QUESTIONS

- Q1-** Describe the conductometric titration of a mixture of strong and weak acid with a strong base?
- Q2-** What are the advantages of Amperometric titrations?
- Q3-** How specific conductivity of solution is determined?
- Q4-** What do you know about Faradaic Current in voltametry?
- Q5-** Give differences between specific conductance and molar conductance?
- Q6-** What is residual current? Give its two components?
- Q7-** What is the role of supporting electrolyte in voltametry?
- Q8-** What is electric arc? Give its uses?
- Q9-** What do you know about electric arc?
- Q10-** What are amperometric biosensors?

LONG QUESTIONS

- Q1-** Explain anodic stripping voltametry?
- Q2-** Describe various kinds of currents contributing to polarographic waves?
- Q3-** Describe the functions of three types of electrodes in classical polarography?
- Q4-** Discuss amperometric titrations along with their advantages?
- Q5-** Discuss various factors affecting conductance of solution?
- Q6-** Write a note on electric spark?

ANALYTICAL CHEMISTRY
(SPECIAL THEORY 1)

2015

SHORT QUESTIONS

- Q1-** Give the types of conductometric titrations?
- Q2-** What are the advantages of electroanalytical techniques over other analytical methods?
- Q3-** What change in conductance is observed when NH_4OH is added to HCl ?
- Q4-** How voltametric techniques are classified?
- Q5-** Give differences between equivalent conductance and molar conductance?
- Q6-** What is the difference between cathodic and anodic current?
- Q7-** What is the role of supporting electrolyte in voltametry?
- Q8-** What is electric arc? Give its uses?
- Q9-** How does electric spark occur?
- Q10-** What are the advantages of polarography?

LONG QUESTIONS

- Q1-** Discuss cyclic voltametry?
- Q2-** Describe various kinds of currents contributing to polarographic waves?
- Q3-** Give advantages and limitations of using DME?
- Q4-** Give applications of amperometric titrations?
- Q5-** Discuss electrodes in Amperometry?
- Q6-** Discuss applications of polarography in inorganic analysis?

ANALYTICAL CHEMISTRY
(SPECIAL THEORY 1)

2014

SHORT QUESTIONS

- Q1-** Define specific and molar conductance?
- Q2-** Write names of some methods used to determine the concentration of unknown solution in classical polarography?
- Q3-** What are the advantages of anodic stripping voltametry?
- Q4-** What is diffusion current? Write mathematical form of Fick's second law?
- Q5-** Draw three types of amperometric titrations?
- Q6-** Describe two methods for measurement of electrolytic conductance?
- Q7-** Differentiate between the arc and spark?
- Q8-** Why mercury electrode is widely used in voltametry?
- Q9-** Explain the importance of supporting electrolyte?
- Q10-** Differentiate between automated and automatic instruments?

LONG QUESTIONS

- Q1-** Describe the working and advantages of dropping mercury electrode?
- Q2-** Describe the factors affecting the conductance of solution?
- Q3-** Write down the advantages of automation process?
- Q4-** Write about the instrumentation for voltametry?
- Q5-** Explain the characteristics of a polarogram?